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CSC360 - Computer Graphics and Digital Image Processing

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Reflection: Computer Graphics & Digital Image Processing

In practice, today's class allowed me to get acquainted with the whole process involved in developing and managing a Java project through Maven, Git, and GitHub. I have learned how to install and configure the Java Development Kit (JDK), check java and javac, as well as configure Apache Maven with the right Java version. In setting up, I came across the Java version compatibility issue, which made me appreciate the significance of properly configuring environment variables and the system PATH.

Next, I created a Maven project and learned about its typical directory structure and the role of pom.xml (Project Object Model). I have learned how groupId, artifactId, version, and Java compiler are specified in the POM file. As an application of theory in practice, I developed a Square Project using Java Swing and AWT. It uses JFrame, JPanel, and Graphics2D to make a graphical window, and draws a square through four separate drawLine() methods, where each line is for a side of the square.

After creating the program, I compiled and built the program using mvn commands like mvn clean package. I understood the process through which the target folder is created and why the files that are created must be ignored using a .gitignore file.

Lastly, I learned the basics of Git and GitHub, including git init, git status, git add, git commit, git clone, and git push. I understood how to clone the course repository and include my Maven project into the repository without disturbing its structure. In summary, this exercise taught me about the entire process starting from setting up Java to creating the Maven project and using Git and GitHub.